
Target-Speaker Speech-to-Text in Overlapping Speech

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Abstract

Modern speech-to-text systems achieve low word error rates on clean, single-speaker audio but degrade sharply when speakers overlap. We present a target-speaker speech-to-text system that, given a short enrollment clip of the desired speaker and a multi-speaker audio mixture, transcribes only the target speaker. Our pipeline combines a frozen pretrained speaker verification model (ECAPA-TDNN) to extract a speaker embedding, a trained speaker-conditioned masking network that isolates the target speaker's spectrogram, and a frozen pretrained ASR model (Whisper) for transcription. We evaluate on synthetic two-speaker mixtures from LibriSpeech at varying signal-to-interferer ratios and overlap ratios, measuring word error rate and character error rate. We show that overlapping speech degrades ASR performance across languages, and present our extraction-based approach to mitigate this degradation.

Attestation of Teamwork

Jibran Iqbal Shah: Designed and trained the speaker-conditioned extraction network, including architecture design, loss function development (power-law compression, hybrid oracle mask loss), hyperparameter tuning, and all related code and writeup.

Yamoah Frimpong Attafuah: Fine-tuned Whisper-small on the WAXAL Akan dataset for Twi speech recognition, set up the WER evaluation scripts and notebooks, and benchmarked both the baseline Whisper and fine-tuned Whisper models across varying overlap ratios and SIR levels for English and Twi. Wrote the code and writeup related to these contributions.

1 Introduction

Modern speech-to-text systems perform well on clean, single-speaker audio but degrade sharply when speakers overlap [Watanabe et al., 2020, Kanda et al., 2020]. This is a common occurrence in meetings, court proceedings, and media such as films or talk shows. Rather than attempting full diarization and separation, target-speaker transcription offers a more focused alternative: given a short enrollment clip, transcribe only the speaker of interest.

Additionally, the rise of full-duplex speech-to-speech models is increasing the demand for labelled transcriptions of natural conversations with interruptions [Défossez et al., 2024, Roy et al., 2026]. A reliable target-speaker system could automatically generate such labels by isolating each participant and producing timed transcripts, even through overlapping speech.

The objective of this project is to develop a target-speaker speech-to-text system that takes (i) a short enrollment utterance of the target speaker and (ii) a multi-speaker mixture, and outputs only the target speaker’s transcript. We test across English and Twi to examine whether the degradation pattern is language-agnostic.

2 Preliminaries and Problem Formulation

Let $\mathbf{x}_{\text{mix}} \in \mathbb{R}^T$ be a single-channel audio mixture containing overlapping speech from $N \geq 2$ speakers. We are also given a short enrollment utterance $\mathbf{x}_{\text{enroll}} \in \mathbb{R}^{T_e}$ of the target speaker s^* . The goal is to produce a transcript $\hat{y}$ that matches the ground-truth transcript y^* of speaker s^* within the mixture, as measured by word error rate $\text{WER} = (S + D + I)/N_{\text{ref}}$, where S, D, I are the number of substitutions, deletions, and insertions and N_{ref} is the number of words in the reference transcript. Note that WER can exceed 100% when the model produces more insertions than there are reference words.

The Short-Time Fourier Transform (STFT) converts a waveform into a two-dimensional complex-valued matrix $\mathbf{X} \in \mathbb{C}^{F \times T}$, where each entry encodes the amplitude and phase of frequency bin f during time frame t . The magnitude spectrogram $|\mathbf{X}| \in \mathbb{R}^{F \times T}$ retains full linear frequency resolution, which is needed to invert back to a waveform after masking via the inverse STFT (iSTFT). We use standard parameters for 16 kHz speech: 25 ms window (400 samples), 10 ms hop (160 samples), FFT size $N_{\text{fft}} = 512$, giving $F = 257$ frequency bins linearly spaced from 0–8000 Hz.

3 Design

Our pipeline consists of three stages, illustrated in Figure 1:

1. **Speaker embedding extraction.** A frozen pretrained ECAPA-TDNN [Desplanques et al., 2020] encodes the enrollment utterance $\mathbf{x}_{\text{enroll}}$ into a speaker embedding $\mathbf{e} \in \mathbb{R}^{192}$.
2. **Speaker-conditioned extraction.** A trained masking network receives the mixture magnitude spectrogram $|\mathbf{X}_{\text{mix}}|$ and the speaker embedding $\mathbf{e}$, and predicts a soft mask $\mathbf{M} \in [0, 1]^{F \times T}$ that isolates the target speaker. The estimated target spectrogram is $|\hat{\mathbf{X}}_{\text{target}}| = \mathbf{M} \odot |\mathbf{X}_{\text{mix}}|$, and the waveform is reconstructed via iSTFT using the original mixture phase.
3. **ASR decoding.** A frozen pretrained Whisper model transcribes the extracted waveform.

Only the extraction network (stage 2) is trained; the speaker encoder and ASR model remain frozen throughout.

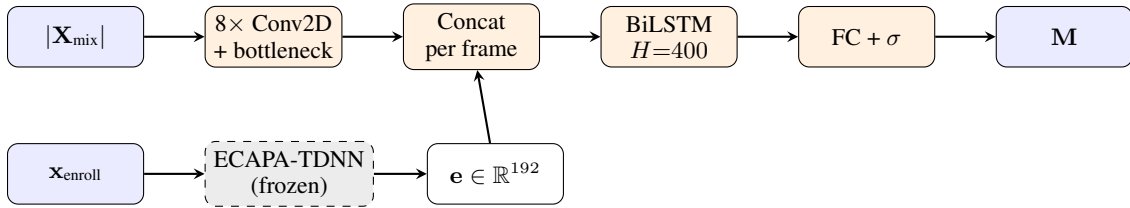


Figure 1: Extraction network architecture. The magnitude spectrogram is encoded by eight 2D convolutional layers (with a channel bottleneck), conditioned on the speaker embedding via concatenation, and processed by a BiLSTM to predict a soft mask $\mathbf{M} \in [0, 1]^{F \times T}$. Dashed box denotes a frozen pretrained model.

4 Methodology

4.1 Speaker embedding

We use a pretrained ECAPA-TDNN [Desplanques et al., 2020] (speechbrain/spkrec-ecapa-voxceleb) to extract a 192-dimensional speaker embed-

ding from the enrollment utterance. The model achieves 0.80% equal error rate on VoxCeleb1 and is kept frozen throughout.

4.2 Extraction network architecture

The extraction network follows the VoiceFilter design [Wang et al., 2019] and has 9.3M trainable parameters.

4.2.1 Convolutional encoder

The magnitude spectrogram is treated as a single-channel 2D image and processed by eight Conv2D layers, each followed by layer normalization (implemented as GroupNorm with one group) and ReLU:

Layer	Kernel (freq $\times$ time)	Dilation (freq, time)	Ch.	Purpose
1	(7, 1)	(1, 1)	64	Frequency-only
2	(1, 7)	(1, 1)	64	Time-only
3–7	(5, 5)	(1, 2^k)	64	Joint, time-dilated ($k=0 \dots 4$)
8	(1, 1)	(1, 1)	8	Channel bottleneck

Layers 1–2 are separable-style convolutions: the (7, 1) kernel scans frequency to detect spectral patterns, while the (1, 7) kernel scans time. Layers 3–7 use joint kernels with dilation increasing only along time (factors 1, 2, 4, 8, 16), giving a temporal receptive field of 31 frames (≈ 310 ms). Layer 8 compresses from 64 to 8 channels.

Unlike 1D convolutions that treat all F bins as independent channels, 2D convolutions share weights across frequency, providing frequency translation equivariance: a harmonic pattern at 500 Hz is recognized by the same filter at 1000 Hz.

4.2.2 Speaker conditioning and BiLSTM

The CNN output $\mathbf{H} \in \mathbb{R}^{B \times 8 \times F \times T}$ is flattened to $\mathbf{g}_t \in \mathbb{R}^{2056}$ per time frame. The speaker embedding $\mathbf{e} \in \mathbb{R}^{192}$ is concatenated at every frame, yielding $\tilde{\mathbf{H}}_t \in \mathbb{R}^{2248}$. A single-layer BiLSTM ($H=400$ per direction) processes the sequence, and two FC layers with ReLU and sigmoid produce the mask:

$$\mathbf{z}_t = \text{ReLU}(\mathbf{W}_1 \mathbf{h}_t + \mathbf{b}_1) \in \mathbb{R}^{600}, \quad \mathbf{M}_t = \sigma(\mathbf{W}_2 \mathbf{z}_t + \mathbf{b}_2) \in [0, 1]^F \quad (1)$$

The BiLSTM is critical because overlap regions are locally ambiguous—a frame where two speakers’ harmonics coincide may only be resolvable given surrounding context in both directions.

4.3 Training objective

4.3.1 Power-law compressed loss

Raw MSE loss is dominated by loud frequency bins (vowels at magnitude $\sim 10,000$) while quiet bins (fricatives at magnitude ~ 1) contribute negligible gradient. Following Wilson et al. [Wilson et al., 2018], we apply power-law compression ($p = 0.3$) before computing MSE:

$$\mathcal{L}_{\text{recon}} = \frac{1}{FT} \left\| (\mathbf{M} \odot |\mathbf{X}_{\text{mix}}| + \epsilon)^p - (|\mathbf{X}_{\text{target}}| + \epsilon)^p \right\|_F^2$$

where $\epsilon = 10^{-8}$ prevents gradient explosion at zero. The same compression is applied to the CNN input to normalize its dynamic range; the mask is still applied to raw magnitudes for reconstruction.

4.3.2 Hybrid loss with oracle mask supervision

The oracle mask is the mask that perfectly recovers the target: $\mathbf{M}_{f,t}^* = \text{clamp}(|\mathbf{X}_{\text{target}}|_{f,t} / (|\mathbf{X}_{\text{mix}}|_{f,t} + \epsilon), 0, 1)$. We found that power-law loss alone allows the

model to find a shortcut—masks biased toward 1.0 that reduce reconstruction error without separating speakers. Adding oracle mask supervision prevents this:

$$\mathcal{L} = \alpha \cdot \underbrace{\|\mathbf{M} - \mathbf{M}^*\|_F^2}_{\text{mask MSE}} + (1 - \alpha) \cdot \mathcal{L}_{\text{recon}}$$

We validated $\alpha = 0.1$ as optimal through controlled overfit experiments (Table 1).

4.4 Data pipeline

4.4.1 Datasets

We use two speech corpora. **LibriSpeech** (English): read speech derived from public-domain audiobooks, 16 kHz mono. We use the dev-clean and test-clean splits. **WAXAL Akan** (Twi): a dataset of Twi speech from the HuggingFace Hub, used to fine-tune Whisper for a language it does not natively support. All audio is loaded at 16 kHz and peak-normalized to $[-1, 1]$.

4.4.2 Synthetic mixture generation

We generate 12,500 two-speaker mixtures from LibriSpeech by summing pairs of utterances from different speakers: $\mathbf{x}_{\text{mix}} = \mathbf{x}_{\text{target}} + \alpha \mathbf{x}_{\text{interferer}}$, where α controls the signal-to-interferer ratio (SIR). We vary $\text{SIR} \in \{0, 5\}$ dB and overlap ratio $\in \{0.1, 0.2, 0.3, 0.4, 0.5, 1.0\}$. The overlap ratio determines what fraction of the target utterance is overlapped by the interferer. Each mixture preserves the aligned clean target waveform for computing oracle masks and ground-truth spectrograms during training.

4.4.3 Train/validation/test split

The 12,500 recipes are split into 10,000 train, 1,250 validation, and 1,250 test samples. Speaker embeddings for all unique enrollment utterances (2,588 embeddings) are precomputed and cached.

5 Numerical Experiments

5.1 Whisper fine-tuning for Twi

Whisper does not natively support Twi (Akan), producing near-100% WER and CER out of the box. We fine-tuned Whisper-small (244M params) on the WAXAL Akan dataset, reducing clean-speech WER from $\sim 100\%$ to 33.3% and CER from $\sim 100\%$ to 12.1%. This fine-tuned model serves as the Twi baseline for our overlap experiments.

5.2 Baseline: overlap degrades ASR across languages

We evaluated Whisper-tiny (39M params) on English and the fine-tuned Whisper-small on Twi across all SIR levels and overlap ratios.

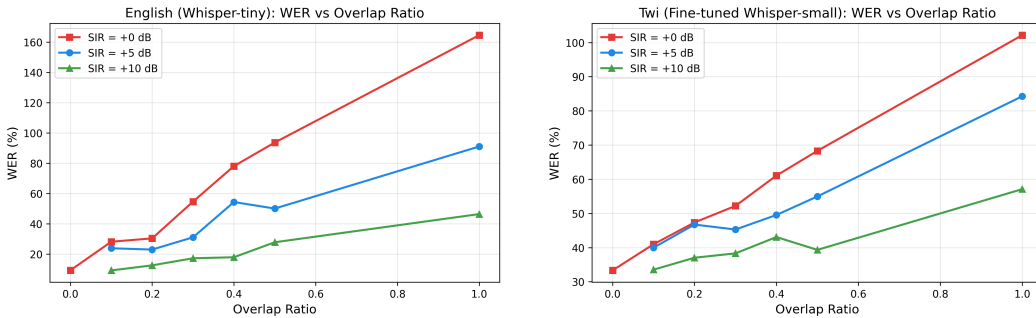


Figure 2: WER vs. overlap ratio for English (left) and Twi (right) at three SIR levels. Both languages show the same degradation pattern: WER increases with overlap ratio and decreases with SIR. At 0 dB SIR with full overlap, English WER exceeds 100%.

English clean baseline: 8.1% WER. At 0dB SIR with full overlap, WER exceeds 100%. Two clean baseline: 34.7% WER with the same degradation pattern. This confirms the problem is language-agnostic.

5.3 Hybrid loss validation

We validated the hybrid loss weight α through controlled overfit experiments (seed=42, 1 batch of 16 samples, 200 epochs, lr= 10^{-4} , compressed input) with dual-metric logging.

Table 1: Hybrid loss ablation on 1-batch overfit test ($p = 0.3$, 200 epochs).

α	Recon PL	Mask MSE	Notes
0.0 (power-law only)	0.0027	0.333	Best recon, but masks diverge
0.05	0.0031	0.0171	
0.1	0.0032	0.0166	Best balance
0.2	0.0033	0.0155	
1.0 (oracle only)	0.0046	0.0184	Good masks, poor recon

The key finding: power-law loss alone ($\alpha = 0$) achieves the best reconstruction metric but produces masks biased toward 1.0 (mask MSE of 0.333 vs. all-ones baseline of 0.158), meaning the model learns to pass everything through rather than separate speakers. Adding 10% oracle mask supervision ($\alpha = 0.1$) reduces mask MSE by $20\times$ with only a 19% increase in reconstruction loss.

5.4 Training

We trained with the validated configuration: batch size 32, learning rate 10^{-3} with ReduceLROnPlateau (factor 0.5, patience 3), Adam optimizer, hybrid loss $\alpha = 0.1$, power-law compression $p = 0.3$, early stopping with patience 7.

5.5 WER evaluation with extraction

We evaluated two trained models on the 1,250-sample test set: the hybrid loss model ($\alpha = 0.1$) and the power-law reconstruction loss model ($\alpha = 0$). Both were trained until validation loss plateaued.

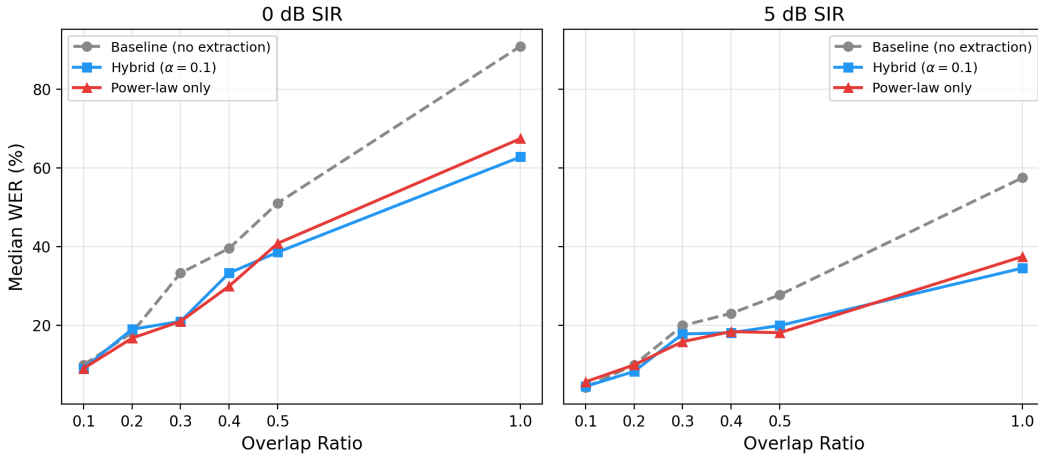


Figure 3: Median WER vs. overlap ratio for baseline (no extraction), hybrid loss ($\alpha = 0.1$), and power-law reconstruction loss only ($\alpha = 0$) at 0 dB SIR (left) and 5 dB SIR (right). Both extraction models reduce WER at overlap ratios ≥ 0.3 , with the largest gains at full overlap.

Both extraction models significantly reduce median WER compared to the baseline at overlap ratios ≥ 0.3 . At full overlap (1.0), the hybrid model reduces WER from 90.9% to 62.8% at 0 dB and

from 57.5% to 34.6% at 5 dB. The power-law model achieves comparable gains (67.4% and 37.5% respectively). Neither model consistently dominates the other: hybrid is slightly better at very high overlap (1.0), while power-law is slightly better at medium overlap (0.3–0.5). At low overlap (≤ 0.2), extraction provides marginal or no benefit, as the interference is brief enough for Whisper to handle without assistance.

6 Discussion

7 Conclusions

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Use of AI

We used Claude Code (Anthropic) as a coding assistant throughout the project for implementing the extraction network architecture, debugging training issues, writing evaluation scripts, and drafting sections of this report. All AI-generated code was reviewed, tested, and modified by the authors. Architecture decisions (hybrid loss formulation, 2D CNN design, hyperparameter choices) were made by the authors based on experimentation and literature review, with AI assistance for implementation.

Consent for Information Sharing

- All group members consent to allow the project materials (report, slides, and presentation recording) to be shared on the course page for future students.

Consent for Use of Project Information in Statistical Analysis

- All group members consent to allow anonymized information about the project (e.g., topic, methods, outcomes, grades) to be used by the instructor for statistical analysis and course improvement.

Group Identification

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